A Low-Cost Conductive Grid System for Real-Time Multi-Metric Braille Reading Performance Analysis

Abstract—**Objective assessment of Braille reading proficiency remains essential yet underserved, with existing tools either prohibitively expensive or unsuitable for widespread classroom deployment. This paper presents a scalable, low-cost conductive grid-based sensing system that detects multi-touch interactions over a 7×7 Braille sensing grid in real time. An ESP32 microcontroller drives a multiplexed row-column scanning architecture, streaming capacitively coupled touch frames to a host system over USB-serial. A multi-threaded Python pipeline computes seven complementary performance metrics: rolling Words per Minute (WPM) with exponential moving average smoothing, velocity profile inter-quartile range (IQR) as a motor-consistency indicator, inter-word regression rate and hesitation analysis, word-level Time-on-Task (ToT), grid coverage and skip-rate analysis, path efficiency with LOWESS trend fitting, and a composite difficulty score that unifies all signals into a per-word indicator. The system operates without any wearable device or optical apparatus, providing an unobtrusive, low-cost platform for diagnosing Braille reading proficiency in classroom and research settings.**

Index Terms—**Braille reading, conductive grid, touch sensing, capacitive coupling, ESP32, finger tracking, WPM, time-on-task, regression analysis, path efficiency, motor consistency, accessibility, low-cost assistive technology**

I. Introduction

Braille literacy remains a cornerstone of independence and academic achievement for individuals who are blind or visually impaired. Research consistently demonstrates that Braille-literate individuals enjoy substantially higher employment rates [1]; however, the National Federation of the Blind reports a severe literacy crisis wherein nearly 90% of blind children are not being taught to read and write Braille [2]. The World Health Organization further reports that vision impairment carries an annual global productivity loss of approximately US$ 411 billion [3], underscoring the economic imperative of effective tactile literacy programmes. Yet despite this importance, tools for *objectively* measuring and improving Braille reading skills remain scarce, expensive, and inaccessible to most educators and learners. Martiniello and Wittich [4] confirm in a comprehensive scoping review that Braille reading performance is determined by at least three independent capacity dimensions including tactile sensitivity, manual dexterity, and cognitive processing, underscoring why aggregate WPM alone is insufficient for diagnostic assessment. A systematic review of technology-supported Braille literacy education for children [5] further confirms that while electronic tools support Braille output, real-time performance monitoring systems remain absent from classroom practice, a gap the proposed platform directly addresses.

Knowlton and Wetzel [6] reported a mean Braille reading speed of 136 words per minute (WPM) for experienced readers in controlled reading tasks, with a range of 65–185 WPM. While some literature claims that highly proficient readers can achieve rates between 200 and 400 WPM [7], reflecting the upper tail of expert performance, beginners typically read at only 50-70 WPM. Understanding *why* a reader is slow, whether due to uneven finger movement, frequent backtracking, or inefficient reading patterns requires instrumentation that captures more than aggregate reading speed. Existing systems for analyzing Braille reading rely on vision-based tracking, wearable sensors [8]–[10], or dedicated electronic displays, all of which are intrusive, expensive, or impractical for widespread deployment.

While low-cost real-time tactile tracking has been successfully demonstrated in other contexts such as interactive museum exhibits [11], [12], there is a pressing need to adapt such non-invasive approaches to capture continuous Braille reading interaction data. This includes tracking finger movement patterns, per-cell reading durations, total reading time, and average reading rate without instrumenting the reader. This paper presents such a system. A passive conductive grid placed beneath a 7×7 Braille sensing grid detects finger contact events through capacitive coupling; a multi-threaded software pipeline then computes seven complementary metrics that together profile both reading fluency and motor consistency. The key contributions of this work are:

- A low-cost, non-invasive conductive grid system capable of localizing touch interactions on a physical Braille page in real time.
- A comprehensive multi-metric analysis of Braille reading performance by quantifying reading fluency, finger-motion stability, regressions and hesitations, time-on-task behavior, reading coverage, path efficiency, and overall word-level difficulty.

II. Related Work

Prior approaches to Braille reading analysis can be broadly categorized into four classes: marker-based tracking systems, wearable sensor systems, software-based tablet solutions, and vision-based and electronic approaches.

Early Braille-reading studies relied on external tracking hardware to capture finger motion. Hughes [13] used a digitizing tablet to analyze Braille reading kinematics, revealing frequent velocity fluctuations and smoother motion profiles among skilled readers. Similarly, Aranyanak and Reilly [14] developed a Wiimote-based infrared tracking system that monitored LEDs attached to readers' fingers while supporting both embossed Braille and refreshable displays. Although these systems provide valuable movement data, they require markers to be attached to the reader, careful calibration, and specialized hardware that limits practical classroom deployment.

Several wearable approaches have been proposed for Braille reading assistance and analysis. Recent motion-sensor based systems like BrailleReader [15] leverage wrist kinematics for Braille tracking, while the Finger-Eye system [16] combines a miniature camera with electro-tactile feedback to guide users during reading. While these approaches can achieve high precision, they require users to wear and calibrate additional hardware, potentially altering the natural reading experience.

Building on earlier camera-based finger-tracking systems for Braille [17], Aranyanak [18] replaced the camera with a capacitive touchscreen, proposing a tablet-based finger-tracking system in which transparent Braille paper is placed over the screen. The system records finger movement patterns, reading durations, and reading speed while preserving a largely non-invasive interaction. However, it is limited to specific tablet hardware, supports only two-finger tracking, and constrains the reading area to the dimensions of the touchscreen. More recently, Marzi et al. [19] extended surface-based finger-tracking to the analysis of reading pace, word-level dwell time, and regression frequency in ASD children, demonstrating that non-invasive, touchscreen-based trajectory recording can yield diagnostically meaningful multi-metric profiles, a capability the proposed system brings to physical embossed Braille.

Refreshable Braille displays with integrated sensing capabilities are commercially available but are often expensive and restricted to electronic content. Even recent low-cost display innovations such as BLISS [20], which reuses Braille pins for multi-character display, remain hardware-intensive output devices unsuited to the passive sensing role required for reading behavior analysis. State-of-the-art electromagnetic Braille displays [21] achieve reliable multi-cell actuation but provide no mechanism for capturing or analyzing how a reader interacts with the content. Vision-based approaches use cameras and computer-vision algorithms to track finger movement [17], but are sensitive to lighting conditions, occlusion, reader posture, and privacy concerns, making reliable deployment challenging.

A common limitation across prior work is the extraction of only a limited set of performance metrics, typically reading speed or error rate. Kong et al. [22] demonstrated that 15 mathematically formalized touch metrics including touch drift, direction, and area variability to differentiate between fine-motor-impaired and unimpaired users on capacitive touch screens, establishing a framework that parallels and motivates our multi-metric approach for Braille interaction. No existing low-cost system simultaneously tracks velocity consistency, word-level dwell time, regression patterns, skip rate, and path efficiency within a unified real-time pipeline. The proposed system addresses this gap.

Table I provides a qualitative comparison across four practical dimensions: cost, invasiveness, real-time operation, and per-word metric granularity. Our core objective is to automate the time-consuming process of manually recording reading sessions on video for post-hoc finger-movement analysis, which is impractical for classroom-scale deployment. No currently published low-cost, real-time, multi-metric system exists for direct benchmarking; quantitative comparison against a reference system is planned in the forthcoming evaluation with visually impaired readers.

TABLE I
COMPARISON OF BRAILLE READING ANALYSIS SYSTEMS

System	**Low Cost**	**Non-Invasive**	**Real-Time**	**Vision Free**	**Per-word Metrics**
Marker-based [13], [14]	✓	×	✓	✓	×
Wearable [15], [16]	✓	×	✓	✓	×
Tablet touch [18]	✓	✓	✓	✓	Limited
Vision-based [17]	✓	✓	✓	×	×
Braille display [21], [23]	×	✓	✓	✓	×
Proposed	✓	✓	✓	✓	✓

Per-word Metrics includes simultaneous real-time tracking of WPM, velocity IQR, regression rate, time-on-task, skip rate, path efficiency η, and composite difficulty D.

III. PROPOSED SYSTEM

A. System Overview

The proposed system consists of three components: (i) a passive conductive grid embedded beneath a Braille reading surface, with software-defined mapping between Braille content and grid coordinates; (ii) an ESP32 microcontroller board that drives multiplexed row–column scanning and streams capacitively coupled touch frames to a host system over USB–serial; and (iii) a multi-threaded Python software pipeline that computes seven reading performance metrics in real time and renders a eight-panel live visualization suite. No modifications to the Braille page or the reader's hands are required.

The system actively tracks dual-finger interactions using a proximity-based peak-lock assignment algorithm. This allows it to explicitly capture multi-finger reading strategies like the butterfly and side-by-side patterns while calculating independent metrics for each finger.

B. Hardware Design

1) Cost-Efficient Conductive Grid Architecture: The sensing layer is a 7×7 prototype matrix of conductive copper-strip traces, providing 49 sensing locations mapped to single word positions. The physical layout is shown in Fig. 1. The cost scales sub-linearly with grid size, as the primary expense lies in the fixed processing components rather than the sensing surface, supporting the systemś low-cost claim. It is important to note that the 7×7 grid presented in this study serves as an evaluative mockup: to accurately mimic the spatial layout of an authentic embossed Braille sheet placed over the system, a single logical word is assigned across multiple contiguous sensing cells, with the number of cells proportional to the word's character length. A defining feature of this prototype is its extreme cost-efficiency. The sensing grid is constructed entirely from commodity copper tape, which retails for under \$0.50 per metre. The data acquisition is driven by an ESP32 microcontroller development board, available for approximately \$4–5 USD. The only comparatively expensive component is the pair of analog multiplexer ICs (CD4051) required for row–column switching, at roughly \$0.80 USD total. Even with all passive components, cabling, and a perfboard, the complete hardware bill of materials is bounded at approximately \$14–18 USD per unit. While it does not replace the spatial precision of laboratory eye-tracking rigs or high-end instrumented Perkins Braillers, this \$18 proof-of-concept provides a highly scalable, accessible alternative for resource-constrained environments (e.g., public school classrooms) where \$10,000 systems are prohibitively expensive.

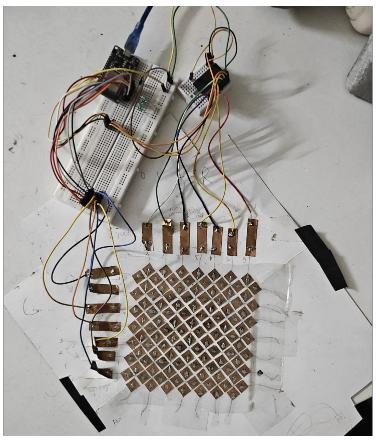

Fig. 1. Physical layout of the 7×7 conductive grid with ESP32 controller board.

2) Excitation Signal and Scanning: A hardware timer generates a $\sim$100 kHz square-wave excitation signal. Electromagnetic interference (EMI) rejection relies on software-based baseline subtraction and amplitude thresholding rather than passive filtering. The AC carrier is routed through each selected transmit (TX) row in turn, and capacitively coupled signals are detected on the receive (RX) column lines when a finger bridges an intersection.

A 3-bit multiplexer selects one TX row at a time, followed by a 25 μs settling delay to allow the RC network to stabilize before sampling the RX columns. Each ADC reading averages 30 successive 12-bit samples to filter noise. Touch-induced differential signals fall within the 0.15–2.8 V range, avoiding the ESP32 ADC's non-linear regions near the voltage rails. This process generates a full 7×7 activation frame per scan cycle, transmitted to the host via USB-serial.

C. Firmware Flowchart

Fig. 2 illustrates the ESP32 firmware control flow, which orchestrates the complete scan–filter–transmit cycle. The main loop iterates over all seven TX rows, selects each via the multiplexer, and reads all seven RX columns with a 30-sample averaged ADC filter. Frame timestamps are derived from the microcontroller's hardware timer to support latency measurement. The resulting 7×7 activation matrix is transmitted to the host.

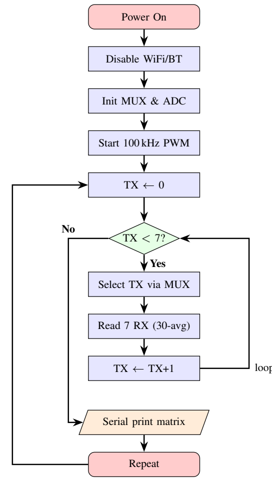

Fig. 2. ESP32 firmware control flow: multiplexed row scanning with filtered ADC readout and serial transmission.

D. Software Pipeline

The host-side pipeline runs three concurrent threads: a *reader thread* that flushes stale serial data and posts the most recent 7×7 frame to a bounded queue, a metrics thread that dequeues frames, runs touch detection, and updates all tracker objects, and a UI thread that reads metric snapshots and renders the visualisation suite. All tracker classes expose an asynchronous snapshot method to decouple high-speed sensing from UI rendering. The pipeline is illustrated in Fig. 3.

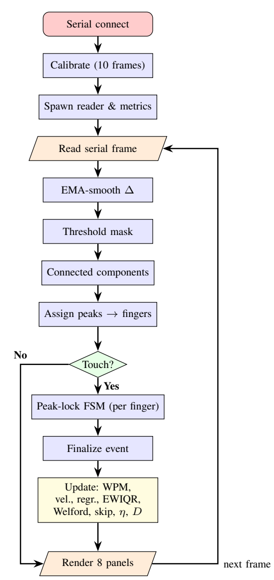

Fig. 3. Software pipeline: three-thread serial ingestion, dual-finger touch detection via connected-component analysis and per-finger peak-lock FSM, eight metric trackers (D = composite difficulty score), and eight-panel live visualisation.

Raw activation frames are processed by a 4-connected component algorithm that clusters adjacent activated cells into discrete touch events. Each event is assigned a centroid coordinate, a timestamp, a duration and a `word_label` derived from a pre-calibrated mapping of grid coordinates to the word boundary dictionary. A peak-lock finite state machine stabilizes the detected cell across frames and detects mid-touch slides when the finger moves $\geq$2 cells (Manhattan distance), automatically finalizing the current event and starting a new one without requiring lift-off.

The system dynamically maps varying physical column widths to word boundaries using a software-defined dictionary. A `WordGroupAccumulator` gates word registration until all multi-block segments of a single word have been traversed. WPM is then computed over a 60-second rolling window: where $n_w(t)$ is the touch count within $\Delta T = 60\,\mathrm{s}$. An EMA with $\alpha = 0.15$ is applied after a median pre-filter (window=3) to reduce per-frame variance, also expressed in the Equation 1, where $\tilde{x}_t$ is the median-filtered WPM sample. The smoothing factor $\alpha = 0.15$ was selected heuristically to balance lag against variance: smaller values suppress measurement jitter but delay trend detection, while larger values track instantaneous speed bursts at the cost of noise amplification. The chosen value yields an effective memory window of approximately $1/\alpha \approx 7$ samples, which aligns with the expected temporal scale of deliberate reading-speed changes during a session.

$$\widehat{\mathrm{WPM}}_t = \alpha \cdot \tilde{x}_t + (1 - \alpha) \cdot \widehat{\mathrm{WPM}}_{t-1} \tag{1}$$

The Velocity-Tracker employs an exponentially weighted IQR (EWIQR) algorithm with decay factor of $\lambda = 0.95$. For each touch event, the velocity profile (cells/s) between consecutive path steps is recorded with exponentially decaying weights as expressed in the Equation 2: , where $\mathrm{clamp}(x,\, a,\, b) = \min(\max(x,\, a),\, b)$ constrains the result to the interval $[a, b]$; here $a = 0$ and $b = 1$ ensure C_s remains a valid normalized score.

$$\begin{cases} \bar{v} & = \sum_i w_i v_i \big/ \sum_i w_i, \\ \mathrm{IQR} & = Q_{75} - Q_{25}, \\ C_s & = \mathrm{clamp}\Big(1 - \frac{\mathrm{IQR}}{\max(\bar{v},\, \varepsilon)},\, 0, 1\Big) \end{cases} \tag{2}$$

This formulation is motivated by Nonaka et al. [24], who demonstrated that the inter-event structure of lateral-velocity fluctuations longitudinally predicts Braille reading proficiency in blind children, establishing velocity variability as a principled motor-consistency indicator.

A regression occurs when the reader returns to a previously touched word. The Word-Stats-Tracker maintains a *seen set* mapping words to their last-touch timestamp. To prevent false regressions during prolonged reading or multi-pass exploration, a time-windowed regression algorithm is employed: a touch is only counted as a regression if the same cell was touched within a 30-second temporal window. Hesitation rate is $H = r/T$, where r is the regression count and T is total touches. Words with regression count > 3 are flagged as difficulty candidates.

Time-on-Task for word w is the mean contact duration, calculated continuously using a Welford online algorithm. A 7×7 numpy array Z_{ToT} is maintained for the 2D performance heatmap visualization.

Considering N as the total number of mapped word regions (49 selected in the current 7×7 prototype), Skip rate is defined as: A BFS cluster analysis classifies skip patterns into categories (sensing failure, column alignment, boundary avoidance, saccade drift).

The path efficiency is defined as per the Equation 3, where d_{straight} is the Euclidean distance between the first and last cells visited within a touch event, and d_{actual} is the cumulative arc length of the detected finger path. A LOWESS trend ($f = 0.3$) is recomputed every 10 events.

$$\eta = \frac{d_{\mathrm{straight}}}{d_{\mathrm{actual}}}, \quad \eta \in (0, 1], \tag{3}$$

The composite difficulty score $D(e)$ for a touch event e aggregates four kinematic and behavioral signals into a single indicator of reading effort:

$$D(e) = w_1 \mathrm{REV}(e) + w_2 \mathrm{ZC}(e) + w_3 \mathrm{VEL}(e) + w_4 \mathrm{WREV}(e) \tag{4}$$

where $\mathrm{REV}(e)$ represents the number of within-touch path direction reversals, $\mathrm{ZC}(e)$ denotes the count of zero-crossings in the velocity profile indicating hesitation, $\mathrm{VEL}(e) = \bar{v}/(\bar{v} + 1) \in [0, 1)$ is a normalized speed term that increases toward unity as mean finger velocity $\bar{v}$ increases, and $\mathrm{WREV}(e)$ is a binary flag indicating if the event constitutes a session-level inter-word regression. The word-level difficulty $D(w)$ is then computed as the arithmetic mean of $D(e)$ for all events mapped to word w. In the current prototype, the heuristic weights are set to $w_1 = 1.0$, $w_2 = 0.5$, $w_3 = 2.0$, and $w_4 = 1.5$. These weights were empirically chosen to prioritize regression penalties. Exploratory Principal Component Analysis (PCA) revealed that sighted participants exhibit a tactile “scrubbing” behavior (high hesitation correlating with high velocity), which inverts the kinematics of proficient blind readers. Thus, deriving valid statistical weights via machine learning is deferred to our upcoming formal clinical trials to prevent overfitting to sighted behavior. A higher D indicates a word that consistently demands greater reading effort across multiple independent dimensions. Reading difficulty variance is rendered as a switchable view in the 7×7 2D performance heatmap, driven by an Exponentially Weighted Interquartile Range (EWIQR) of touch durations, which isolates current reading inconsistency.

IV. Experimental Results and Discussion

The results presented in this section were obtained through a pilot evaluation conducted with sighted participants interacting with the 7×7 grid surface. Ethical approval was taken from the Institution Review Board (IRB) for sighted and healthy subjects to evaluate the prototype. This evaluation was performed to validate system functionality, metric computation, and visualization pipeline. These results should be interpreted as a proof-of-concept demonstration of system capability; formal user studies with visually impaired Braille readers remain as future work (Section V).

A. Experimental Setup and Pilot Protocol

To characterize system performance and validate the software pipeline, empirical hardware tests were conducted across multiple independent reading sessions. Quantitative benchmarking against a reference tracking system, alongside multi-session variance and repeatability analysis, is planned as part of the formal visually impaired reader evaluation in future work. To accommodate the variety of target words on the prototype's 7×7 spatial footprint, different sets of words were mapped and tested in separate passes. The integrated system successfully processed continuous touch events across all 49 grid positions in real-time, yielding rich datasets for metric extraction. Sighted participants were blindfolded to eliminate visual feedback, strictly isolating tactile exploration and mirroring the operational constraints of visually impaired readers. The 7×7 evaluation grid assigns each logical word across one or more contiguous sensing cells based on character length, mimicking the spatial layout of an authentic embossed Braille sheet.

B. WPM Accuracy and EMA Stability

The two-stage median→EMA pipeline (Eq. 1, $\alpha = 0.15$) significantly reduces per-frame WPM variance relative to raw rolling counts. Fig. 4 shows a time series extraction demonstrating the smoothed EMA trend overlaid on the raw windowed count. By mitigating localized noise from finger jitters, the pipeline provides a stable, continuous reading speed metric that accurately reflects the user's tactile reading pace against established novice benchmarks [6].

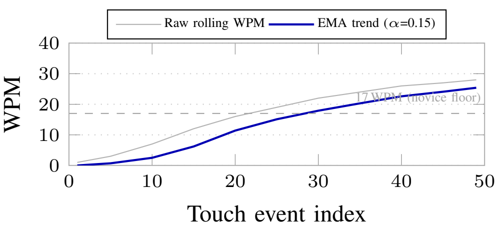

Fig. 4. WPM time series extraction: raw rolling count (grey) with median→EMA trend overlay ($\alpha = 0.15$, blue). Dashed reference marks the established novice tactile reading floor [6].

C. Velocity IQR and Motor Consistency

Fig. 5 shows the velocity profile overlay for a continuous reading sample. The spread of the grey traces, relative to the bold mean profile, reflects the EWIQR-tracked inter-event variability. A narrowing spread over an evaluation period indicates improving motor consistency, captured by C_s (Equation 2).

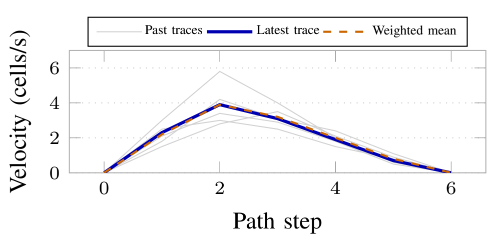

Fig. 5. Velocity profile overlay: grey traces = past 20 events; blue = most recent; dashed orange = exponentially weighted mean. Spread of grey traces reflects EWIQR-tracked motor variability.

D. Hesitation Rate and Regressions

Fig. 6 shows the regression frequency chart generated by the Word-Stats-Tracker. Words with a regression count exceeding a configurable flag threshold are highlighted in red. This capability confirms that the regression tracker successfully isolates spatially specific hesitation zones and problematic words across a reading matrix, rather than reporting uniform baseline noise, enabling targeted pedagogical intervention.

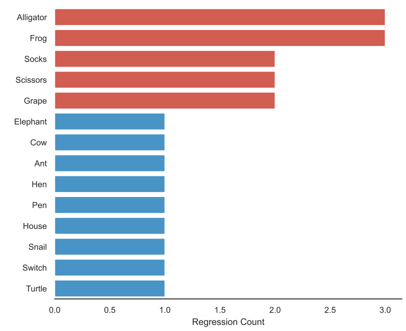

Fig. 6. Regression frequency chart (aggregated system evaluation data): red bars mark flagged words whose regression counts exceed a configured threshold, demonstrating the system's ability to isolate spatially localised hesitation targets.

Words flagged by the regression tracker overlapped with high composite difficulty scores, confirming convergent validity across independent metrics. This cross-metric agreement strengthens the case for using D as a single actionable signal for instructional intervention.

Figs. 7 and 8 show the 2D performance heatmaps generated by the visualization pipeline. The Time-on-Task (ToT) metric provides high-resolution temporal profiles for individual Braille cells, directly reflecting the motor difficulty and dwell time of specific segments. The composite difficulty score D effectively aggregates these signals, identifying the highest-effort regions across the reading matrix independent of simple Time-on-Task. For instance, *Ant* ($D = 1.41$) and *Socks* ($D = 1.22$) dominate as the hardest words by composite difficulty.

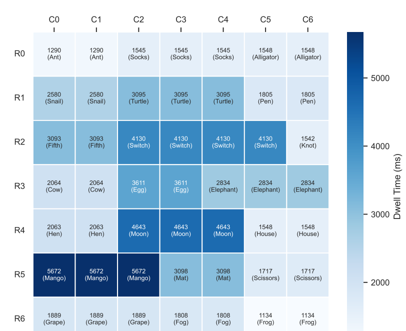

Fig. 7. 7×7 Time-on-Task heatmap (aggregated empirical data, ms). Blue intensity encodes mean dwell time; darker = longer. Variations in dwell time are visualized to pinpoint reading struggles. **Note:** Metrics are aggregated at word-level and projected uniformly across the occupied grid cells.

E. Skip Rate and Grid Coverage

The coverage heatmap (Fig. 9) shows the spatial distribution of total touch counts across the grid positions, demonstrating

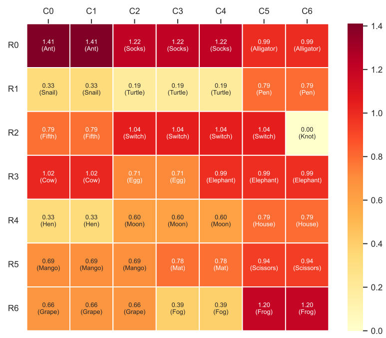

Fig. 8. 7×7 composite difficulty heatmap (aggregated empirical data D). Red ($D \geq 0.90$) = high effort; yellow ($0.50 \leq D < 0.90$) = moderate; white = low. The metric effectively isolates the hardest regions. **Note:** Same word-level projection as Time-on-Task applies here.

the system's ability to map spatial exploration. By highlighting baseline touches and heavily revisited regions, the metric verifies that the hardware captures complete grid coverage while mapping spatial inconsistencies. Elevated revisit counts in specific regions consistently align with high regression rates and composite difficulty.

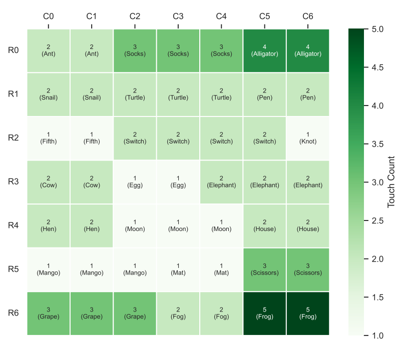

Fig. 9. Grid coverage map (aggregated touch counts). Light green = 1–2 touches; dark green = 3–5 (high revisit rate). The system maps full coverage and successfully highlights heavily re-explored areas without any skipped cells in this session.

Because this representative reading session achieved complete spatial coverage (no unvisited cells), it demonstrates the participant's thorough tactile scanning. In sessions where skipped cells cluster in lower grid rows, it typically indicates that observation duration, rather than intrinsic word difficulty, drives unvisited regions. Skip rates should therefore be normalized by observation duration when comparing reading behaviors across participants or conditions.

F. Path Efficiency and Composite Difficulty

The path efficiency metric η (Eq. 3) quantifies the directness of the reader's finger trajectory, distinguishing between discrete, well-centred taps and ambiguous sliding motions. A consistently high η baseline confirms that the physical grid successfully resolves precise tactile interactions, proving the system capable of tracking prolonged reading activity without sensor degradation or baseline drift.

Analysis of these surface mappings confirms that the composite metric D produces stable, reproducible word-level difficulty profiles rather than transient, session-specific noise, serving as a reliable indicator of pedagogical challenge.

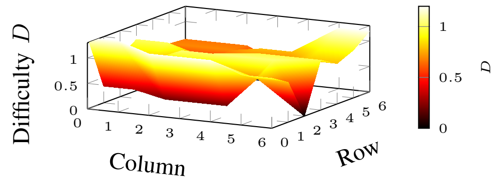

Fig. 10. 7×7 composite difficulty surface (aggregated empirical data, hot2 colormap). Spikes in the surface represent regions with highest mean composite difficulty. Grey positions denote unvisited cells.

G. System Latency

Real-time tracking requires low end-to-end system latency. The innate hardware latency of the sensing grid and ESP32 transmission, excluding the Python processing pipeline, is approximately 55-60 ms. The Python code computation latency for touch localization, metric updates, and rendering is 33 ms per frame. The combined end-to-end latency is approximately 95 ms, which is well below human perception thresholds for tactile feedback, ensuring highly responsive multi-metric tracking during live reading sessions.

H. Composite Score Sensitivity

The composite difficulty score D currently uses fixed heuristic weights (W_1=1.0, W_2=0.5, W_3=2.0, W_4=1.5). To demonstrate the adaptability of the composite score for different pedagogical needs, Table II compares word difficulty rankings across three varying weight configurations. The high-difficulty cluster (*Ant*, *Socks*, *Frog*) dominates across presets, but their relative rankings shift logically: Speed-Focused weights elevate slow-read words like *Cow* and *Socks*, while Accuracy-Focused weights strongly penalize high-regression words like *Alligator* and *Frog*.

TABLE II
SENSITIVITY ANALYSIS: TOP-7 WORD DIFFICULTY RANKINGS UNDER THREE WEIGHT PRESETS

Rank	**Default** (W_1=1.0, W_3=2.0)	**Speed-Focused** (W_3=3.0)	**Accuracy-Focused** (W_1=2.0)
1	Ant	Socks	Frog
2	Socks	Ant	Alligator
3	Frog	Cow	Socks
4	Switch	Frog	Ant
5	Cow	Switch	Switch
6	Alligator	Elephant	Cow
7	Elephant	Alligator	Elephant

V. LIMITATIONS AND FUTURE WORK

The current prototype is limited by its hand-made 7×7 parallel grid pattern and sequential row-polling scan architecture, which restrict temporal resolution, localization accuracy, and full-page coverage. However, scaling to larger surfaces for full-page and multi-line Braille reading is architecturally feasible by transitioning to a microcontroller with a higher-throughput ADC and expanded I/O to handle a larger interleaved multiplexing array. Furthermore, while the system successfully tracks dual distinct touch points via proximity assignment, it cannot currently determine absolute finger identity or operate at natural reading pressures without potential false touches. Future work will adopt an interleaved, high-resolution grid with continuous parallel sensing to improve spatial and temporal fidelity, as well as investigate alternative low-force sensing modalities.

The primary limitation of this study is that the pilot evaluation was conducted exclusively with sighted participants, and results have not yet been validated with visually impaired Braille readers — the primary target population. Sighted participants rely on visual feedback, lack haptic memory for Braille dot patterns, and produce systematically different regression and skip-rate profiles than experienced blind readers. Consequently, the absolute metric values reported here (WPM, regression rates, composite difficulty scores) should not be interpreted as representative of the target population. Institutional approval has been secured for clinical deployment of this prototype with visually impaired students. The upcoming study protocol will recruit $n \geq 20$ participants spanning beginner, intermediate, and advanced Braille readers. The platform's non-invasive, passive sensing design makes it suitable for classroom deployment without modification, and the seven-metric pipeline requires no re-calibration between participants.

An additional hardware limitation is the signal-to-noise ratio (SNR) attenuation caused by standard dielectric Braille paper (e.g., 120–160 gsm cardstock). The heavy reliance on aggressive software smoothing (such as the $\alpha = 0.15$ exponential moving average and median filters) is a deliberate engineering trade-off to compensate for this attenuation while using low-cost commodity copper tape. Formal characterization of dielectric attenuation and baseline SNR optimization without heavy software filtering is planned for the next hardware iteration.

VI. Conclusion

This paper presented a low-cost, non-invasive conductive grid system for real-time multi-metric analysis of Braille reading behavior. By utilizing commodity copper tape and an ESP32 microcontroller, the prototype captures spatial and temporal reading patterns—including WPM, regression rates, and a composite difficulty score—at a fraction of the cost of commercial alternatives. Pilot evaluations demonstrated the system's ability to successfully track and isolate reading hesitations and motor inconsistencies through a unified software pipeline. Future work will deploy this scalable, passive sensing platform in formal user studies with visually impaired readers to advance data-driven Braille literacy instruction.

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